

YASH SOLANKI

AI-driven Application Engineering | Functional Application Management

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PROFILE SUMMARY

Application Support Engineer with 3+ years of experience across enterprise CRM and OSS/BSS platforms, specializing in root cause analysis (RCA) and SLA-driven operations, including enterprise CRM support for Swisscom and OSS/BSS incident support for BSNL, India's national telecom operator. Hands-on with Python, RAG, embeddings, and MCP-based tool integration for GenAI-assisted diagnostics and ticket triage.

CORE SKILLS

Application Support: L2/L3 Application Support, Root Cause Analysis (RCA), SLA Compliance, MTTR Reduction, Escalation Management, Role-Based Access Control (RBAC)

Telecom Domain: OSS/BSS, CRM

Areas of Expertise: Order-to-Activate (O2A), Provisioning & Activation

AI/LLM Engineering: Python, RAG, MCP, Agentic Workflows, Tool Calling, Embeddings, Vector Search, Vector Databases (pgvector), Structured Output Generation, Evals, Guardrails/AI Safety Validation, FastAPI, Groq, sentence-transformers

AI-Driven Application Intelligence: LLM-Driven RCA Automation, Ticket Triage & Routing, Duplicate/Similarity Detection, Text Classification, Human-in-the-Loop AI Design, Cross-Module Dependency Mapping, GenAI-Assisted Diagnostics

Databases & Operating Systems: MySQL (Data Validation), PostgreSQL, Linux

Cloud & Enterprise Platforms: Microsoft 365 Administration, Azure AD (User Access & Sync Management), Docker

API Testing & Enterprise Tools: Postman (REST/SOAP APIs), JIRA, Zendesk, BMC Remedy

Delivery & Methodologies: Release Deployment, Functional/Flow Testing, Defect Management

PROFESSIONAL EXPERIENCE

Application Support Engineer – CRM Application Support & AI-Driven Operations (OSS/BSS)

FGS B.V. (Client: Swisscom) | Rijswijk, Netherlands | Jun 2026 – Jul 2026

- **GenAI-Assisted Diagnostics & Automation:** Deployed Claude and Python scripting into CRM incident investigation workflows for the Swisscom account, speeding up log parsing and cutting down manual documentation effort during day-to-day support.
- **SLA & Incident Ownership:** Managed day-to-day incident resolution across Microsoft 365 / Azure AD, resolving user access issues, workflow failures, and sync errors while consistently meeting SLA targets.
- **AI-Driven Ticket Intelligence:** Applied duplicate-ticket detection via embeddings on the Swisscom CRM/OSS-BSS queue, using retrieved matches to speed up RCA lookup during daily triage, with output checked against evals/guardrails before use in live ticket workflows.
- **AI Research & Independent Tool Building:** Serving on FGS's Research team for the Swisscom engagement, identifying operational problems and independently designing and building working AI-driven solutions using free, market-available tools, including an automated detection-to-resolution pipeline covering AI log classification, RCA generation, similarity search, and SLA-tracked escalation.
- **Gen-AI Analytics & Agentic Workflow Deployment:** Supported FGS's early-stage AI in a Box / CX platform rollout on the Swisscom engagement, contributing to agentic workflow design for governed text and data analytics.

Senior Engineer – AI-Driven Operations & Product Support (OSS/BSS)

Whale Cloud Technology India Pvt Ltd | Hyderabad, India | Apr 2025 – Jan 2026

- **SLA & Incident Management:** Improved end-to-end Order-to-Activate (O2A) throughput and reduced workflow exception queues across a large-scale subscriber environment by resolving CRM-to-OSS state-transition blocks, collaborative waiting failures, and masterless order discrepancies through structured incident management and system refiring.
- **Data Integrity & Backend Validation:** Maintained data integrity between CRM and OSS by executing targeted MySQL database queries to resolve backend discrepancies, correcting spatial parameters across affected service instances, and driving cross-vendor bulk migrations of active directory numbers across core telecom platforms.

- **AI-Driven Ticket Intelligence:** Built the RAG retrieval workflow itself (embeddings, duplicate-ticket matching, MCP-exposed tool calls feeding LLM-based RCA generation), reducing redundant R&D escalations by giving engineers prior-fix context before routing.
- **System-Level Intelligence:** Used cross-module dependency mapping across CRM, OSS, BSS, Billing, DSCM, Mediation, and Network Management during incident triage to trace failures back to their originating module faster.

System Engineer – L2/L3 OSS/BSS & Application Support

Tenet Computer and Communication Pvt Ltd | Bengaluru, India | Oct 2022 – Mar 2025

- **Incident Routing, MTTR & Escalation Management:** Accelerated critical fault resolution and reduced Mean Time to Resolve (MTTR) across BBNW, FTTH, and NGN modules by re-engineering L1-to-L3 workgroup mapping and ASSIGN_GROUP routing logic (Zendesk, BMC Remedy), while preventing unnecessary L3 escalations through resolving L1 authentication faults and managing high-priority B2B client communications directly.
- **Platform Security & RBAC:** Secured system access for telecom operators across the South Zone by executing strict Role-Based Access Control (RBAC) policies within Zsmart OSS and eliminating duplicate user profiles to strengthen platform security.
- **SLA Compliance & Analytics:** Maintained strong SLA compliance across major tech hubs (Bengaluru, Pune, and Hyderabad) by analyzing daily and weekly operational intelligence reports (MIS, Pending Dockets, Average Ticket Duration) to systematically clear L2 bottlenecks.
- **Linux Log Analysis & RCA:** Minimized production downtime caused by critical backend exceptions through deep root cause analysis (RCA) on "Next Tache" calculation errors, parsing application logs within Linux environments to ensure high availability across microservices-based distributed systems.
- **API Validation & Infrastructure Config:** Mitigated critical interface integration delays at the ISAP and OFM portals by executing REST/SOAP API payload analysis in Postman, while eliminating operational provisioning bottlenecks across DSLAM/LMG network nodes.
- **Delivery & Defect Management:** Ran functional flow testing across BBNW, FTTH, and NGN modules ahead of releases, logging and tracking defects in JIRA to keep day-to-day rollouts separate from the platform-wide BSNL upgrade work covered under Key Projects.

KEY PROJECTS

Automated Incident Detection & Resolution Pipeline

Independent Project | Feb 2026 – Present

- **Detection Layer:** Built a service that continuously monitors for failures across three signal types (alert manager triggers, data reconciliation mismatches, and automated test failures) and forwards each as a structured alert.
- **AI Classification Layer:** Built an AI service that classifies each incoming alert by category and confidence score, turning raw alert text into a structured triage result in seconds instead of manual review.
- **AI Resolution & RCA Layer:** Built an AI incident-management service that auto-diagnoses root cause with evidence, searches for similar past incidents and prior fixes, tracks SLAs with real-time auto-escalation, and auto-assigns and notifies engineers and customers, closing the loop with no manual handoff.
- **Unified Workbench & Intake:** Automated email-to-ticket intake with auto-classification and SLA start, surfaced alongside RCA and similar cases in a single engineer-facing screen for zero context-switching.
- **Feedback Loop:** Connected resolution outcomes back into the detection layer, so every resolved incident sharpens future classification accuracy.
- **Stack:** Python 3.11+ and FastAPI backend, PostgreSQL with pgvector for similarity search, Groq JSON-mode LLM for RCA reasoning, sentence-transformers (all-MiniLM-L6-v2) embeddings, and Docker Compose deployment on fully free-tier infrastructure.
- **Full-Chain Integration:** Owned end-to-end integration across all three services: one alert, three stages, zero manual handoff, from detection to resolution.

AI-Assisted Telecom Ticket Intelligence & Escalation Platform

Built via Whale Cloud Technology | Apr 2025 – Jan 2026

- **Retrieval & Reasoning Pipeline:** Extended the retrieval pipeline with multi-step reasoning for evidence-grounded resolution recall, not just matching similar tickets but chaining retrieved evidence into a structured triage decision, with human override retained on every routed case.

- **Quality & Observability:** Built an eval suite gating LLM output before production use, with guardrails for safety validation on auto-generated RCA/triage decisions and full tracing and debugging across the reasoning pipeline.
- **Cross-Module System Intelligence:** Designed and built the dependency-mapping layer itself, modeling relationships across CRM, OSS, BSS, Billing, DSCM, Mediation, and Network Management to surface downstream failure impact and recurring weak points ahead of time.
- **Platform Capability Expansion:** Introduced retrieval, reasoning, and system-analysis capabilities not previously present in the platform, extending it from a manual R&D escalation portal into an AI-assisted intelligence layer for telecom incident handling.

Zsmart 9.2 Platform Upgrade | BSNL (Government of India)

Executed via Whale Cloud Technology & Tenet | Oct 2022 – Jan 2026

Delivered continuously on the same BSNL client engagement across two vendor partners as part of a standard IT services staffing transition.

- **Cutover & Release Stabilization:** Executed live cutover support and production validation for the Zsmart 9.2 upgrade across BSNL's national subscriber base, maintaining system availability and minimizing planned migration window overruns.
- **Data Integrity & Database Validation:** Secured post-release stabilization by executing targeted MySQL queries to audit backend data consistency, identifying and resolving critical database migration discrepancies before they impacted active billing cycles.
- **System & Flow Validation:** Guaranteed seamless order activation across network layers by validating end-to-end business flows and parsing system logs to resolve state-transition blocks within the modernized platform architecture.
- **Functional Testing & Defect Management:** Executed rigorous functional test cycles across impacted modules (OSS/BSS, CRM, DSCM), isolating complex platform bugs and systematically tracking defects in JIRA to ensure resolution before major deployment rollouts.
- **Core Network Integration:** Managed the operational integration of the Huawei CDOT voice module during the upgrade cycle, conducting real-time provisioning checks to ensure zero downtime for existing active directory lines.

EDUCATION

B.E. – Computer Science

Rajiv Gandhi Proudyogiki Vishwavidyalaya, Bhopal | 2016 – 2020